Temperature, Pressure, and Ocean Depth at Dana Point Harbor

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***Abstract*— This paper details the development and experimental testing of an autonomous underwater vehicle (AUV) constructed as part of the Experimental Engineering Course (ENGR080) at Harvey Mudd College. The AUV was deployed at Dana Point Harbor, California to measure water temperature, pressure, and ocean depth on April 24th, 2026, from 8:00 am to 10:00 am in order to generate a topographical map of the ocean floor. It was concluded that the AUV successfully measured temperature from 20 ± 0.50°C to 17.59 ± 0.43°C , robot depth (using pressure) from 0.0 ± 0.2 m to 1.2 ± 0.3 m , and ocean depth from 1.65 ± 0.15 m to 2.13 ± 0.13 m at 95% confidence. For future work, a more accurate GPS, longer and more dense surface paths, and more interactions could allow a more detailed ocean map to be created.**

***Index Terms*—Autonomous Underwater Vehicle (AUV), Depth Mapping, Sonar, Marine Robotics, Experimental Engineering, Dana Point**

I. INTRODUCTION

The objective of this project is to develop an Autonomous Underwater Vehicle (AUV) capable of analyzing temperature, pressure, and ocean topography depth independently. Monitoring temperature is important, because it tells the scientific community how the atmosphere and ocean surface interact, which is extremely relevant in the context of climate change [17]. Furthermore, data regarding pressure and topographical depth are utilized by marine chemists and biologists to create models that calculate currents, tides, and predict hazards like coastal flooding and rip tides [16]. In fact, Bathymetry, or underwater topography, is used to validate the safety of water vessel navigation, responsible for aligning shipping routes [18]. However, the collection of quality bathymetric data is often limited by nautical charts as they prioritize navigational safety over the topography of the seafloor itself. [18] In regions like the Baltic Sea, data like this is also classified for operations like navies, which can result in low-resolution datasets for marine researchers interested in understanding the shape of the ocean floor [18]. To overcome these challenges, this project is focused on integrating temperature, pressure, and Sonar for topography at a low-cost (~$50) robot build to allow for independent autonomous mapping of the ocean floor. In this way, researchers can better understand the underwater environment to protect ocean life.

II. SENSOR SELECTION

A. Temperature

For temperature measurements, a thermistor was chosen. A thermistor is a sensor that changes its resistance based on the temperature in the environment around it [2]. The relationship between the resistance across the thermistor and the temperature in the environment is governed by The Steinhart-Hart Equation:

$$T = (A_1 + B_1 ln\, R + C_1 ln^2\, R + D_1 ln^3\, R)^{-1} \quad (1)$$

A thermistor was chosen over alternative methods due to its availability in the lab, low cost, accuracy for its readings, and moderate response times [2]. The specific thermistor that was chosen was the Murata NXFT15WB473FA2B150, which is a Negative Temperature Coefficient (NTC) thermistor, meaning an increase in temperature decreases its resistance [4]. The equation that defines resistance from the datasheet [4] is shown below:

$$R = R_0 exp(B(\frac{1}{T} - \frac{1}{T_0}) \quad (2)$$

Where R_0 is 47kΩ, the resistance at 25°C, T_0 is 25°C, and B is a constant that changes based on the ranges you are working with. The operating temperature range is between -40°C and 125°C, which provides a large range to make measurements. During the month of April, the water temperature at Dana point was reported to be from 15°C to 16°C [1], so to allow for a large margin of error, the thermistor interface circuit was designed to accommodate a temperature range between 10°C to 19°C.

B. Pressure

For the robot to dive, there needed to be some P control to confirm the right depth of the robot. For these diving depth measurements, an absolute pressure sensor was selected. An absolute pressure transducer converts any and all applied pressure into a voltage signal that can be analyzed. The relationship between the applied pressure and the corresponding output voltage is defined by the following function in the datasheet: [3]

$$V_{out} = 5V * (0.0012858 * P + 0.04) \ (3)$$

where P is the applied pressure in kilopascals (kPa).

An absolute pressure sensor was chosen over alternative methods due to its availability in the lab, linear relationship from pressure to voltage output, and ease of integration. The specific pressure sensor that was chosen was the MPX5700ASX, which is a silicon piezoresistive transducer provided to measure absolute pressure from 0 kPa to 700 kPa, which is very suitable for fluid systems like Dana Point [3]. It also operates successfully over a temperature range of 0°C to 85°C [3], which fits the needs for the expected temperature range at Dana Point Harbor. For the purposes of the project, the plan was to allow a dive from 0 meters to 4.5 meters.

For relating pressure to depth, the hydrostatic pressure equation was used, which states that:

$$P_{absolute} = P_{atmosphere} + \rho gh \quad (4)$$

absolute pressure is equal to the pressure in the atmosphere, summed with the product of ρ, the density of the fluid, g, the gravity on earth, and h, the depth from the surface. The projected pressure at Dana Point was 101.3 kPa [1], and considering that ρ $\approx 1020\, kg/m^3$ for salt water and $g \approx 9.81\, m/s^2$ on earth, a 4.5 meter dive would result in 146.3 kPa of pressure, and this is within the pressure sensor operating range.

C. Sonar (Sound Navigation And Ranging)

For the harbor depth measurements, a Sound Navigation and Ranging (or Sonar) based approach was implemented using a sound signal. 'Time of Flight' Sonar works by transmitting a sound pulse from a speaker with a known frequency and pulse length. This sound pulse travels through the water, and then is reflected off of the ocean floor, as well as other objects that are in the water. Then, it is picked up by the second component, which is the microphone. Then, measuring the time delay between the transmission of the signal and the reception of the reflected signal. The time of flight is then used to calculate distance [11], via Equation 5:

$$D = C\frac{\Delta T}{2} \ (5)$$

where D is the distance in meters, C is the speed of sound, and ΔT is the time interval from transmission to reception. For the speed of sound, the Del Grosso equation was considered, stating that:

$$C = C_{000} + \Delta C_T + \Delta C_S + \Delta C_P + \Delta C_{StP} \ (6)$$

Where C_{000} = 1402.392 m/s, while ΔC_T, ΔC_S, ΔC_P, and ΔC_{StP} are changes in this base speed of sound due to temperature changes, salinity, depth, and StP (Standard Temperature and Pressure). [11] For this project, it was assumed that the change of salinity is minimal for the depths of the ocean that were to be mapped, so temperature and pressure sensors were selected.

For the purposes of transmission, a 555 was used to generate the desired frequency to an LM384 5W Audio Power Amplifier, and finally the signal pulse was played using a Waterproof Exciter (or speaker). The 555 timer was implemented due to easy integration, and to limit any complicated software from the teensy to drive certain signals at the right frequency. The LM384 audio amplifier was selected to drive the speaker due to its large output voltage gain, and minimal total harmonic distortion for its range of frequencies [5]. Figure 1 shows a Bode plot of the output voltage gain plotted against frequency, showing that the Audio Amplifier starts with a gain of around 39dB, and drops down as the frequency reaches the order of Mhz.

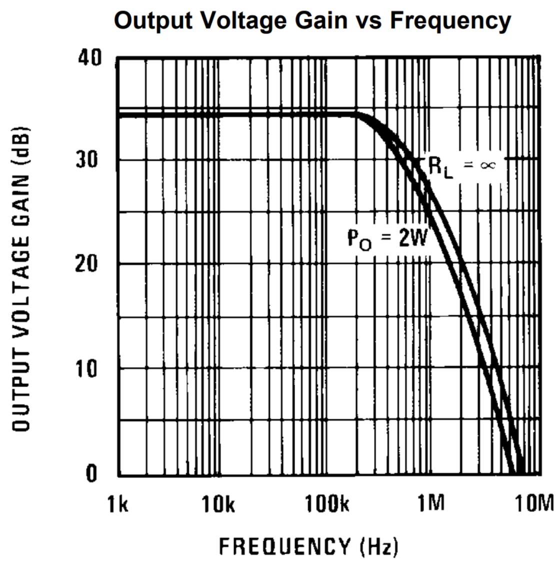

Fig. 1. Audio Amplifier Bode Plot [5]

This is important, as the audio amplifier should produce high gain for the speaker to be as loud as possible. This helps the echo remain a noticeable signal to be received from the microphone. In regards to the speaker, the DAEX25W-8 was implemented for its minimal impedance and high sound pressure level [12]. Figures 2 and 3 show the Gain plots for the Exciter. Since the speaker had the largest dBSPL (Sound Pressure Level) at 3kHz and a minimal impedance of around 9ohms, the 3kHz signal was selected.

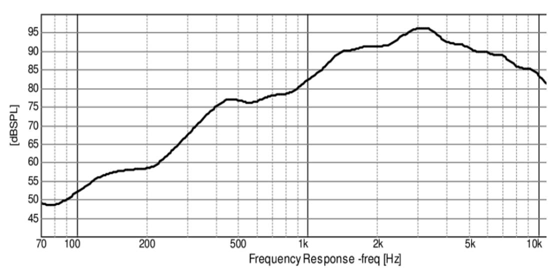

Fig. 2. Gain Plot for DAEX25W-8 Exciter [12]

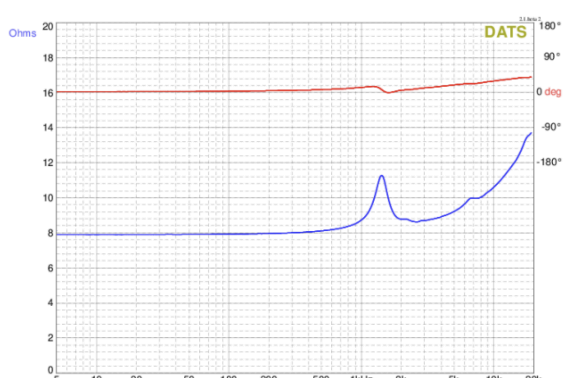

Fig. 3. Gain & Phase Plot for Impedance [12]

III. MECHANICAL & ELECTRICAL DESIGN

A. Mechanical Design & Integration

In order to effectively collect data on the terrain of the ocean floor, the robot needed to minimize rotation and tipping during surface deployments. As a result, a high center of buoyancy by placing the pool noodles at the top of the frame was utilized, while the mass will be concentrated lower down to ensure that the AUV naturally rights itself. The wider frame will also increase stability. There are four horizontal motors, two in the z-direction and two in the x and y directions each. This will allow us to make horizontal movement without the AUV rotating about the z axis, which will allow us to keep better track of where the robot is located and its orientation at any point in time. The design was chosen for its ability to dive quickly while also maintaining as much stability as possible on the surface. A winch was attached to fishing line for team recovery of robot, in case of unforeseen circumstances. Figures 4 and 5 show the sketches and final result.

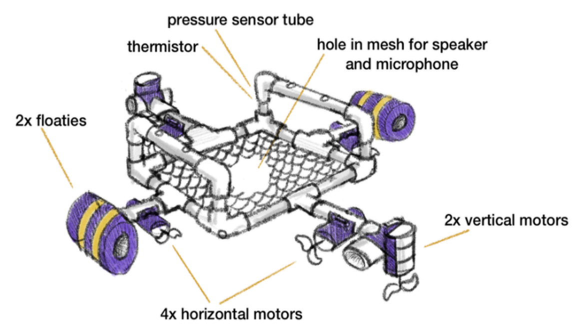

Fig. 4. AUV Sketch

Fig. 5. Final AUV Design

A. Thermistor Interface Circuit

As stated in Equation 2, the thermistor relates temperature to its resistance, so to turn that into a voltage which the Teensy can measure, a voltage divider was created, with the thermistor as the resistor connecting 5V power to the second resistor. With a range of 10°C to 19°C (or 283.15K to 292.15K), the expected resistance from the thermistor would be from 96kΩ to 62kΩ (Equation 2). For the voltage divider, a 75kΩ resistor was used, meaning an input voltage of 5V results in 2.19V to 2.73V across the temperature range. While this fits into the full Teensy input range of 0 to 3.3V [13], expanding the range of the output voltage could allow for more accuracy and resolution in the temperature measurements. This can be done by utilizing an offset amplifier circuit, as shown in Figure 6 below:

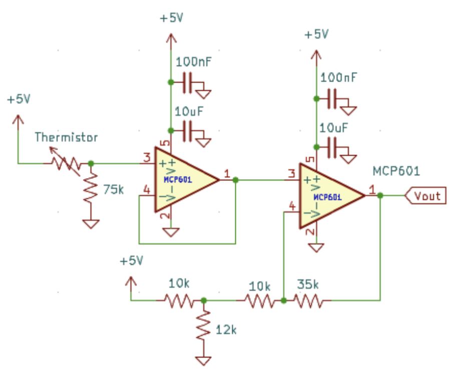

Fig. 6. Thermistor interface circuit

The circuit begins with a unity gain buffer, to prevent loading the thermistor sensor. The output is then used as an input to a non-inverting offset amplifier. The equation that defines such a circuit is shown below in Equation 7:

$$(1 + \frac{R_f}{R_{n1}})V_+ - \frac{R_f}{R_{n1}}V_{n1} = V_{out} \quad (7)$$

This results in a gain of 1.98V, and an offset of -3.5V. The resulting range from 10°C to 19°C is 0.38V to 2.83V, which fits well into the range of the Teensy. The 5V is created by a LM7805 linear voltage regulator from the 12V battery. This same voltage regulator is used for the pressure circuit as well. Given that the output of the temperature sensor is effectively a DC voltage, there are no concerns on the 10Hz sampling rate from the Teensy [13]. To integrate the thermistor with the rest of the AUV, the thermistor was electrically taped and heatshrinked, and was packaged into a penetrator bit with other electrical connections. The thermistor probe itself was taped to the side of a PVC tube of the robot, ensuring that the thermistor was stable and far from the motors.

B. Pressure Interface Circuit

As stated in Equation 3, the pressure sensor relates pressure to its voltage output. With a range of 101.3 kPa to 146.3 kPa, the expected output voltage for the depth of 0 to 4.5 meters would be 0.855V to 1.15V (Equation 3). While this fits into the full Teensy input range of 0 to 3.3V, expanding the range of the output voltage could allow for more accuracy and resolution in the pressure measurements, and would make the P control for the dive more exact. This can be done by utilizing an offset amplifier circuit, as shown in Figure 7:

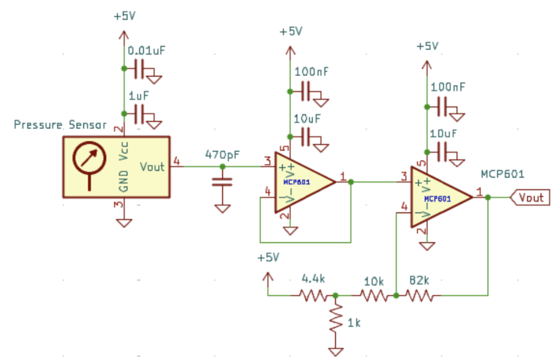

Fig. 7. Pressure interface circuit

The circuit begins with a unity gain buffer, to prevent loading the pressure sensor, and includes bypass capacitors to ensure a stable output. The output is then used as an input to a non-inverting offset amplifier, which works as presented in Equation 8. This results in a gain of 9.2V, and an offset of -1.52V. The resulting range from 0 meters to 4.5 meters is 0.317V to 2.84V, which fits well into the range of the Teensy. To integrate the pressure sensor with the rest of the AUV, a corresponding tube was epoxied into a penetrator bolt, was rolled in a circle, and oriented downwards, so no water would enter the AUV's electrical box. Female headers were attached to the circuitry board, in case the pressure sensor were to be damaged and could be replaced with a functioning one.

C. Sound and Ranging Circuit

The Sonar system is a critical component for depth mapping, consisting of a Transmit (TX) chain to generate an acoustic pulse and a Receive (RX) chain to process the reflected signal from the harbor floor.

Transmit (TX) Chain:

The TX chain utilizes an NA555 timer in astable mode to generate a continuous square wave [9]. This signal is subsequently amplified by an LM384 5W Audio Power Amplifier to drive the waterproof speaker. The frequency and duty cycle of the pulse are determined by the timing resistors R_1 and R_2, and the timing capacitor C. Based on the astable configuration, the frequency and duty cycle are calculated below [9]:

$$f_{out} = \frac{1.44}{(R_1 + 2R_2)C} \qquad (8)$$

$$Duty\ Cycle\ (\%) = \frac{R_1 + R_2}{R_1 + 2R_2} * 100\% \qquad (9)$$

By choosing resistor values of $R_1 = 820\Omega$, $R_2 = 2k\Omega$, and a capacitor value of $C = 0.1uF$, a 2.987 kHz signal, and a duty cycle of 58.5% are generated. The output from the 555 timer is then passed through a voltage divider and bypass capacitor, to stabilize a 0.5V input voltage (which is required for the audio amplifier). Then, the LM384 Audio Amplifier drives the speaker, which is able to play the correct frequency. The schematic of the circuit is shown in Figure 8, depicting the 555 timer and audio amplifier:

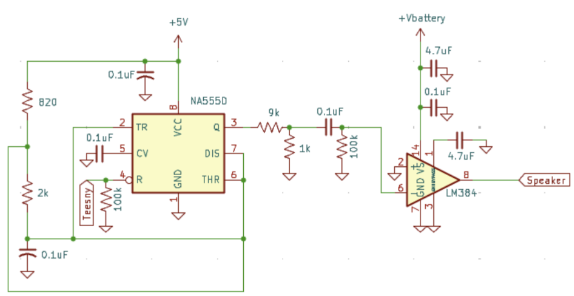

Fig. 8. Transmit (TX) Sonar circuit

To control the operation of this transmit circuit, the Teeny and a $100k\Omega$ pulldown resistor were connected to the reset pin of the 555 timer, and by coding the Teeny to send a "HIGH" signal, the 555 timer can be programmed to send pulses at a set pulse width and duration.

Receiver (RX) Chain:

To receive the signal from the speaker, an interface circuit for the CME-1538-100LB microphone was created. This includes a setup as required from the datasheet, a 10,000 (or 80dB) Gain stage, a Band Pass filter centered at 2.987 kHz, and a Maximum detector circuit with a time constant of $\tau = 20ms$. The schematic of the receiver chain for this circuit is shown in Figure 9.

The receive chain for the Sonar measurements were designed to amplify the extremely small voltage signals captured by the microphone, while filtering out unnecessary noise. First, the datasheet setup for the electret microphone was implemented. The purpose for this setup is to filter out unnecessary noise from the microphone.

This was followed by 4 non-inverting amplifiers. The gain for these amplifiers were defined by the following equation:

$$Gain = 1 + \frac{R_f}{R_g} \qquad (10)$$

where R_f is the feedback resistor connecting V_{out} to V_- and R_g is the ground resistor connecting V_{out} to ground. Given that this circuit uses a resistor combination of $R_f = 90k\Omega$ and $R_f = 10k\Omega$, the gain per stage is 10, which in four stages leads to a total gain of 10,000.

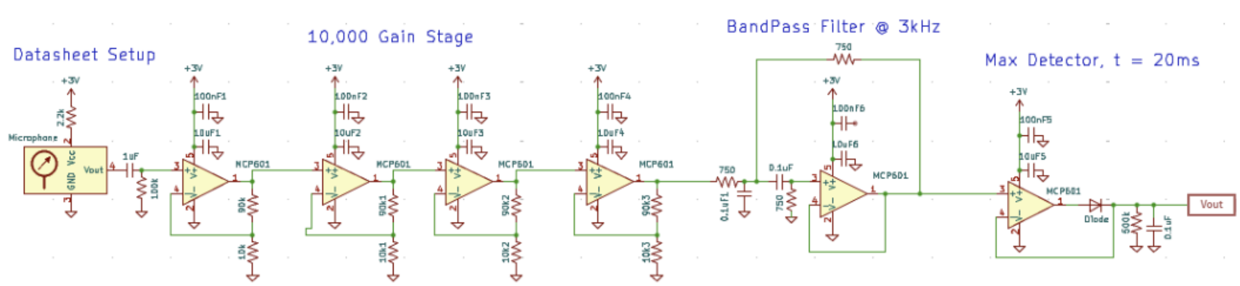

Fig. 9. Receive (RX) Sonar circuit

Considering that there are huge gains and high frequencies for this circuit, the Gain Bandwidth-Product (or GBP) must be considered. The GBP of an operational amplifier is defined by the product of the gain and bandwidth [10]. Therefore, the bandwidth is the GBP divided by the gain of the Op-Amp. The MCP601 that was used for this circuit has a GBP of 2.8 MHz, meaning that for a gain of 10, there is a bandwidth of 280 kHz. The goal is to have the bandwidth larger than the frequency of the 2.987 kHz signal, so that the signal can be read by the Teensy. However, the actual bandwidth does in fact cascade for each gain stage. The equation that relates the bandwidth of the total system to the bandwidth of each stage [11] is the following:

$$BW_{total} = BW_{stage}\sqrt{2^{1/n} - 1} \qquad (11)$$

where BW_{total} is the total bandwidth across the 4 Op-Amps, and BW_{stage} is the bandwidth of just 1 stage, and n is the number of stages. Even with this large shrinkage in bandwidth, the circuit ends up with 121.8 kHz of bandwidth, which is much larger than the roughly 2.987 kHz signal. This is why the gain was distributed across four stages, to mitigate the effects of Gain-Bandwidth Product.

The next stage of the receive chain is the Bandpass Filter. Using a Sallen-Key filter, a second-order band pass filter was created to isolate the 2.987 kHz signal, and ignore the noise of other signals and noises [6]. To find the corner frequency for such a filter, the following equation was used:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{R_f + R_1}{C_1 C_2 R_1 R_2 R_f}} \qquad (12)$$

For $R_f = R_1 = R_2 = 750\Omega$ and $C_1 = C_2 = 0.1\mu F$, the center frequency is 3001.05 Hz. Similarly, the Q-factor can be solved to find the relationship between Bandwidth and frequency, where BW = Q*f [6]:

$$Q = \frac{\sqrt{(R_1 + R_f)R_1 R_f R_2 C_1 C_2}}{R_1 R_f (C_1 + C_2) + R_2 C_2 (R_f - \frac{R_b}{R_a} R_1)} \qquad (13)$$

(Note: $R_a = 0$ and $R_b = \infty$ for this case) This results in a Bandwidth of 1414.67 Hz. A bode plot for this bandpass filter is plotted in Figure 10. By implementing this bandpass filter, the circuit can attenuate any high or low frequency noise, and focus on the 2.9 kHz signal that was sent with the speaker.

The final component of this receive chain is the max detection circuit. This works by utilizing a diode, capacitor, and resistor on the output voltage of the circuit. The op-amp compensates for the voltage drop in the diode. The RC circuit is responsible for generating a sawtooth-like signal, where the voltage will jump when the speaker first emits the sound, and then will spike a few milliseconds after from the echo. The charge will slowly dissipate through the large resistor, until the echo arrives, charging up the capacitor once more. By comparing the time difference between those two peaks, it is possible to calculate the distance using the speed of sound. The time constant for the max detect circuit is found by multiplying $R * C$, which in this case is 20 milliseconds. This time constant represents the amount of time for the max detector to discharge from peak voltage to 0V.

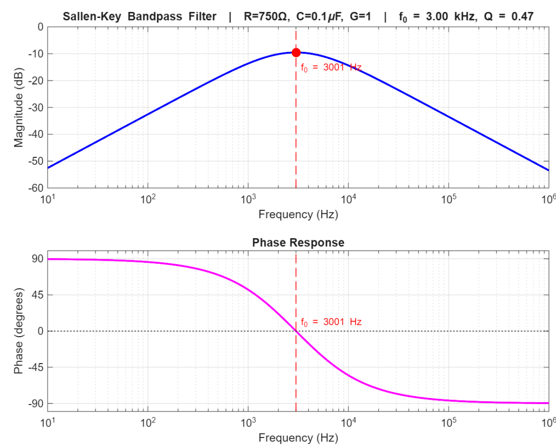

Fig. 10. Bode Plot for Bandpass Filter

One final concern for this circuit is the frequency. The Nyquist-Shannon sampling theorem states that the frequency that is measuring must be more than double the frequency of the signal that is being measured [14]:

$$f_{sampling} > 2f_{measured}$$

However, the Teensy mounted on the Motherboard only samples at 10Hz. To accomplish this, the BurstADC code was implemented. This arduino code samples the desired analog pins at 7.1kHz, and saves the data after the burst is completed, in a datalog file. By utilizing the BurstADC sampler, the Nyquist-Shannon sampling theorem is satisfied, as 7.1kHz > 2*2.987 kHz. Stitching the bursts from BurstADC allows the Teensy to find the two peaks from the max detector.

IV. SENSOR CALIBRATION

Thermistor

Five water baths were used to calibrate the thermistor by measuring the temperature of each bath and its respective output voltage. The results are shown in Figure 11:

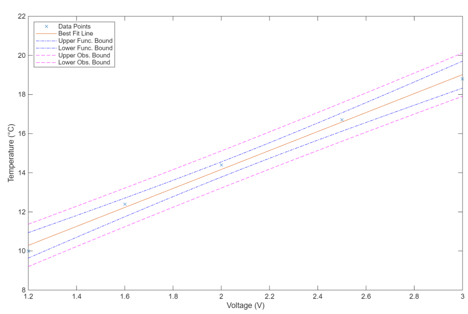

Fig. 11. Calibration Curve for Temperature

Pressure/Depth

The MPX5700ASX pressure sensor was calibrated by measuring the output voltage at known depths using a graduated cylinder with clear water and a meterstick. The resulting calibration curve demonstrated a strong linear relationship between voltage and depth, confirming the reliability of the sensor for P-control during diving maneuvers.

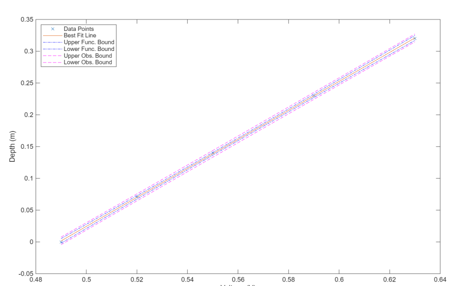

Fig. 12. Calibration Curve for Depth

Sonar (TX & RX)

The Sonar system was calibrated by taking measurements at fixed, known distances from a flat floor in water. By comparing the time delay between the transmitted 2.987 kHz pulse and the received echo, the mean for a set distance was found, and was used to alter the post-processing needed to determine the Time of Flight and the corresponding Depth. Thresholds were also placed for the height of the peaks, to ensure that random noise was not recognized as an "echo". An example of a calibration run is shown in Figure 13:

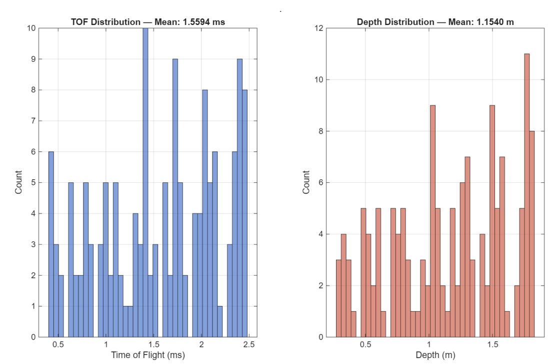

Fig. 13. Calibration Curve for Speaker & Microphone

V. DEPLOYMENT DETAILS

The AUV was deployed many times at the Dana Point Pier & Bay. The testing protocol involved uploading either surface-tracking or depth-control code to the Teensy microcontroller.

Surface Protocol

For surface-tracking deployments, each trial was recorded by first initiating the motherboard and connecting the Teensy to a laptop, to upload the necessary code and view the Serial Monitor. This was to ensure that the GPS module on the Motherboard was confirmed to lock on satellites. When the GPS locked, the computer was disconnected from the Teensy, and the waterproof box was sealed, and secured to the AUV using velcro. The code had a small buffer of 30 seconds to allow for the robot to be thoroughly checked before letting it run its projected path in the water.

The surface protocol focused solely on the Sonar measurements, while the diving protocol focused on temperature and pressure (depth) readings. During the runs, the team followed the AUV in kayaks to monitor its progress and ensure safe recovery. Data was logged to an onboard SD card and reviewed after each run to verify the integrity of the sensor readings. Figure 13 shows the path taken by the robot for the Surface. The intended path was a 3 meter by 3 meter box, however currents and other crafts interfered along the way.

Dive Protocol

For dives, the protocol was relatively the same, except temperature and pressure measurements were made instead. These measurements took place at the Dock, located to the left of the GPS surface path that is displayed in the following Figure 14:

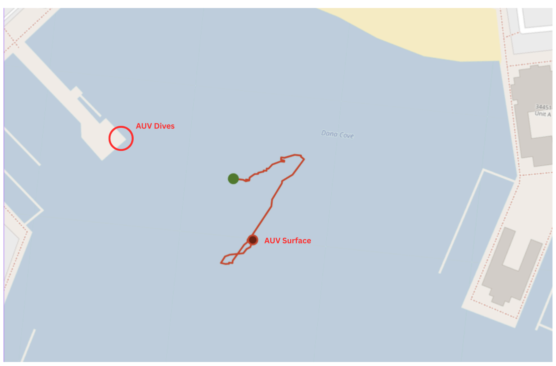

Fig. 14 Surface Path & Dock for Dives [15]

VI. RESULTS & DISCUSSION

Upon receiving the data, the first notice was that the thermistor temperature data was extremely noisy. To filter the data properly, a moving average was applied to the dataset, which is shown in Figure 15. The AUV successfully measured temperature from 20 ± 0.50°C to 17.59 ± 0.43°C before short-circuiting, which is believed to be caused by water conductivity shorting the two thermistor pins, causing the Op-Amp to rail out. The results were within the ground truth measurement of the water temperature from a digital thermometer, which was reported to be 18.5 ± 0.1°C at the surface. Then, temperature was also analyzed as a heat map laid over a depth vs. time graph, shown in Figure 16. As the robot dives into the water, there seems to be a slow decrease in the temperature, with noise causing less continuous data than expected. As for the P control, the robot reached the desired depth of 1 meter, and remained in that spot as instructed. This shows effective use of the pressure sensor as a method of diving.

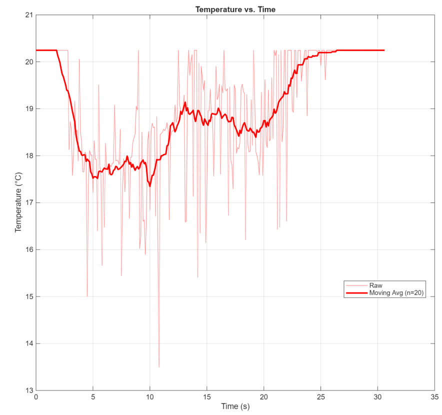

Fig. 15. Moving Average for Temperature Data

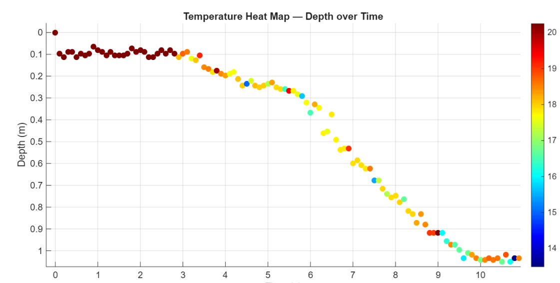

Fig. 16. Temperature Heat Map over Depth vs. Time.

The robot dropped from 0.0 ± 0.2 m to 1.2 ± 0.3 m. This was checked with a pressure tube meter stick, as well as Fishing and Nautical Charts data, showing that the depth where the robot ended was indeed 1.2 meters at 9:00 AM, when the dive was recorded. Note that the slope and intercept that was calibrated had to be adjusted, due to the salt water density in the ocean. This changed the pressure results, and these adjustments were made in the calibration code. As for the Sonar mapping, the datalog burst files were stitched together, and were analyzed to find the peak echo after the speaker was played for each burst. By lining up vectors for the Latitude, Longitude, and the resulting distance for each burst, it was possible to find the discrete distances of the ocean floor. These discrete measurements were splined into a 3D plot, shown in Figure 17. From the plot, the AUV successfully plotted depths from 1.65 ± 0.15 m to 2.13 ± 0.13 m at 95% confidence.

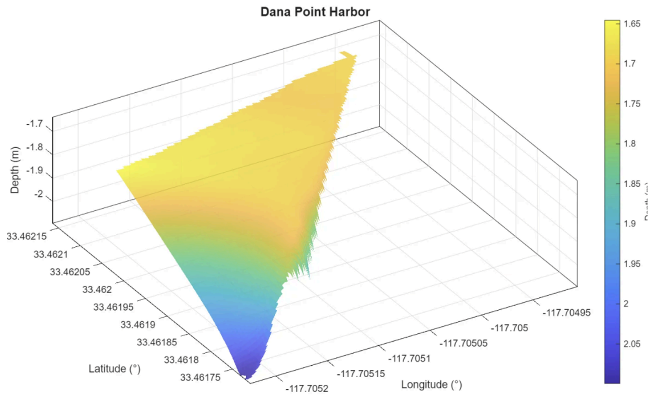

Fig. 17. Topography Map of Path along Dana Point Harbor

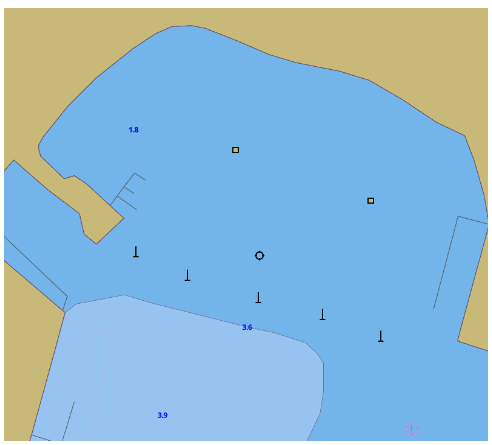

Fig. 18. Ground truth for Dana Point Harbor Depths (Discrete, not Continuous) [7]

Since this was in the middle of the harbor bay, a simple meterstick cannot be used as a ground-truth measurement, especially since the data that was reported had depths of more than 1 meter. As a result, the Fishing and Nautical chart website was used as a ground truth measurement, shown in Figure 18. One important note is that the Fishing and Nautical chart data was very discrete, with a report of 1.8 m covering the entire front of the bay, and 3.6 m being reported after the buoys. Given these discrete measurements, the depth plot seems fairly accurate, given that the measured depths of 1.65 and 2.13 meters were between the discrete bounds of 1.8 meters and 3.6 meters on the website. From these results, it seems that the AUV was successful in its mission to plot a topography map of its path.

VII. CONCLUSION & FUTURE WORK

Overall, the project demonstrated that an autonomous vehicle equipped with a low-cost thermistor, pressure transducer, and custom Sonar circuitry can effectively map underwater environments. While the AUV met its primary goal of depth mapping, several areas for improvement were identified:

1) Implementing higher-density sensor sweeps would enhance the resolution of the topographic maps, especially in areas with complex seafloor geometry. This would allow for more accurate data, as well as larger pools of data for other researchers to integrate into their studies.

2) Improving horizontal and depth control algorithms would reduce noise caused by drift, making data collection more repeatable. The surface and dive control were solely P control, and by implementing other methods such as PI or PID controllers, the robot could traverse the paths much more carefully and successfully.

3) Future iterations should focus on improved waterproofing and insulation for the thermistor to prevent short-circuiting under high-current conditions. While the thermistor seemed to have worked in the lab and with previous tests, upon deployment it easily short-circuited and caused the interface circuit to rail out.

4) And finally, a more accurate GPS module would result in better surface control for the AUV, and would minimize time between each run (waiting for the GPS to lock on satellites). This would also allow for extremely precise GPS locations, and would enhance the surface control for the AUV.

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REFERENCES

[1]“Dana Point Water Temperature (Sea) and Wetsuit Guide (CAL - Orange County, USA),” *www.surf-forecast.com*. https://www.surf-forecast.com/breaks/Dana-Point/seatemp

[2] “Lab 4: Temperature Sensors – HMC E80,” *Github.io*, 2025. https://hmc-e80.github.io/labs/lab4/

[3] “Freescale Semiconductor Data Sheet: Technical Data.” Available: https://www.nxp.com/docs/en/data-sheet/MPX5700.pdf

[4] “NTC Thermistors R44E.pdf,” 2020. Available: https://www.globtek.com/pdf/manual-datasheets/NXFT15WB473FA1B050.pdf

[5] “LM384 5W Audio Power Amplifier Check for Samples: LM384 1FEATURES.” Available: https://www.ti.com/lit/ds/symlink/lm384.pdf

[6] Wikipedia Contributors, “Sallen–Key topology,” *Wikipedia*, Nov. 02, 2019. https://en.wikipedia.org/wiki/Sallen%E2%80%93Key_topology

[7] “Fishing & Nautical Charts Web App,” *Gpsnauticalcharts.com*, 2026. https://fishing-app.gpsnauticalcharts.com/i-boating-fishing-web-app/fishing-marine-charts-navigation.html?title=DANA+POINT+HARBOR+boating+app#17.09/33.46197/-117.70451 (accessed May 08, 2026).

[8] “Reference Designs – HMC E80,” *Github.io*, 2026. https://hmc-e80.github.io/reference/design/ (accessed May 08, 2026).

[9] “xx555 Precision Timers.” Available: https://www.ti.com/lit/ds/symlink/se555.pdf?

[10] Cadence PCB Solutions, “Gain-Bandwidth Product,” *resources.pcb.cadence.com*, Sep. 21, 2023. https://resources.pcb.cadence.com/blog/2023-gain-bandwidth-produc

[11] H. Gemmrich, “Speed of sound in water: what it is and how it’s measured,” RBR, May 05, 2023. https://rbr-global.com/speed-of-sound-in-water/ t

[12] Dayton Audio, “Dayton Audio - DAEX25W-8 Waterproof 25mm Exciter 10W 8 Ohm,” Daytonaudio.com, 2026. https://www.daytonaudio.com/product/1181/daex25w-8-waterproof-25mm-exciter-10w-8-ohm (accessed May 08, 2026).

[13] PJRC, “Teensy® 4.0,” *www.pjrc.com*. https://www.pjrc.com/store/teensy40.html

[14] Wikipedia Contributors, “Nyquist–Shannon sampling theorem,” *Wikipedia*, Nov. 14, 2019. https://en.wikipedia.org/wiki/Nyquist%E2%80%93Shannon_sampling_theorem

[15] Google Maps, “Google Maps,” Google Maps, 2021. https://www.google.com/maps/place/Dana+Point (accessed May 08, 2026).

[16] N. O. and A. A. US Department of Commerce, “How is bathymetric data used?,” *oceanservice.noaa.gov*. https://oceanservice.noaa.gov/facts/bathyuses.html

[17] “The importance of ocean monitoring,” *Voice of the Ocean*, Oct. 12, 2022. https://voiceoftheocean.org/importance-of-ocean-monitoring/

[18] B. Hell, B. Broman, L. Jakobsson, M. Jakobsson, Å. Magnusson, and P. Wiberg, “The Use of Bathymetric Data in Society and Science: A Review from the Baltic Sea,” *AMBIO*, vol. 41, no. 2, pp. 138–150, Oct. 2011, doi: https://doi.org/10.1007/s13280-011-0192-y.